

FieldComms IMS: An Open-Source, Offline-First Incident Management Platform for Amateur Radio Emergency Communications

Grant Proposal to Amateur Radio Digital Communications (ARDC)

Applicant organization	McHenry County Emergency Services Volunteers (MCESV) — 501(c)(3) public charity — club callsign K9ESV — serving McHenry County Emergency Management Agency (MCEMA), Illinois
Project lead	Jim Rospopo, KE4CON — FieldComms system architect and developer
Amount requested	\$15,439
Project duration	12 months
Grant category	Amateur Radio — emergency communications, open-source software release, and community replication
Open-source commitment	Software: GNU AGPLv3 (ARDC-preferred) · Documentation: CC-BY-SA 4.0 · Public repository: github.com/KE4CON/FieldComms-IMS

PROJECT SUMMARY

FieldComms IMS is a complete, working, volunteer-built incident management platform for amateur radio emergency communications. Two Raspberry Pi 5 servers, a Wi-Fi 7 mesh, and dual cellular/satellite uplinks host 32 browser-based tools — net control logging with offline FCC callsign lookup, a full five-section ICS platform, APRS tactical mapping, Winlink integration, and a dedicated AMPRNet (44Net) gateway — all functioning with zero internet and no app installation on any operator device. Version 1.0 is feature-complete with over 190 pages of professional documentation.

This grant funds three things: (1) the production hardware deployment for McHenry County RACES/ARES/Starcom operations, replacing the developer-owned prototype; (2) the formal open-source release of all software and documentation under AGPLv3 and CC-BY-SA 4.0 licenses, packaged so any club can replicate the system; and (3) a structured outreach program — hamfest presentations, a build workshop, and a step-by-step replication guide — to help other emergency communications groups build their own FieldComms systems.

ALIGNMENT WITH ARDC GOALS

Doing. FieldComms puts modern digital tools in the hands of every operator at every net and every activation. It moves a county emergency communications program from paper logs to a synchronized digital operation spanning amateur radio, Winlink, JS8Call, APRS, and public-safety interoperability — and it gets used weekly, not just in disasters.

Learning. Weekly amateur and Starcom check-in nets run on the same tools used in activations, so every net is a training session. The 190+ pages of open documentation — installation guide, user manual, field quick reference, and structured beta test protocol — teach Linux, networking, Python services, and RF integration to every volunteer who builds or operates one.

Experimenting. The platform is a living testbed: volunteers extend it with new tools (the animated NWS radar page and WAN failover monitor were added by volunteers this year), and the open codebase invites clubs everywhere to adapt it.

44Net integration. FieldComms includes a dedicated AMPRNet gateway Pi that routes 44.0.0.0/8 for every device on the deployed network, with FCC-callsign authentication and Part 97 access logging. Every FieldComms deployment is a new, actively used 44Net node — directly advancing the network ARDC stewards.

Precedent. This proposal follows the pattern of ARDC-funded projects such as Oregon HamWAN (emergency digital network expansion), Bay Area Mesh (open-source resilient networking for responders), and the Haiti International Friendship ARC emergency network — combining emergency communications capability with open technology and a replicable model.

THE PROBLEM

When disasters disable commercial infrastructure, most emergency communications volunteer groups revert to paper logs, whiteboards, and hand-carried ICS forms — because the digital tools they use day-to-day assume working internet. Commercial incident management platforms cost \$10,000-50,000 per year, are closed-source, and fail precisely when infrastructure fails. No free, open-source, offline-first alternative packaged for club deployment currently exists.

WHAT FIELDCOMMS ALREADY IS

FieldComms v1.0 is complete and working — this is not a proposal to begin development. The system comprises 26 web applications, 15 Python services, 17 systemd units, and installation automation, developed over more than 800 volunteer hours:

CAPABILITY	DETAIL
Net control logging	Amateur (ARES/RACES) and Starcom loggers · offline FCC lookup of 800,000+ licensees · automatic ICS-309 export · net inactivity alarm
Incident Command System	All five ICS sections · live T-card resource board · Planning P cycle · ICS-213/214/309 forms with print output
Situational awareness	Offline APRS tactical mapping (Graywolf + YAAC merged) · animated NWS radar · live weather alerts · HF propagation
Radio integration	Winlink (Pat + Winlink Express import) · JS8Call · APRS RF · IC-7300 station integration
44Net / AMPRNet	Dedicated gateway Pi · WireGuard tunnel · callsign-authenticated status dashboard · Part 97 access log
Resilience	RAID 1 mirrored storage · dual-carrier cellular primary WAN · Starlink automatic failover · full offline operation · battery/generator power
Documentation	190+ pages: 56-page installation guide, 56-page user manual, 43-page user guide, field quick reference, 24-page beta test protocol

PROJECT PLAN AND TIMELINE

PHASE	MONTHS	ACTIVITIES AND DELIVERABLES
1 — Production build	1-3	Procure and assemble production hardware per bill of materials. Execute the 24-page beta test protocol end-to-end. Deliverable: operational production system owned by MCESV.
2 — Open-source release	2-4	Apply the GNU AGPLv3 license to all software and CC-BY-SA 4.0 to all documentation. Publish complete repository with installation automation. Write the FieldComms Replication Guide — a step-by-step build document for other clubs. Deliverable: public v1.0 release.
3 — Exercise validation	4-7	Deploy at a McHenry County emergency management exercise and at weekly nets (amateur and Starcom). Collect operator feedback via the beta protocol; fix and re-release. Deliverable: exercise after-action findings published to the repository.
4 — Outreach & replication	6-12	Present FieldComms at two regional hamfests and one Illinois ARES/RACES forum. Host one hands-on build workshop for neighboring county teams. Publish a recorded system walkthrough video (CC-BY). Deliverable: at least one neighboring organization beginning its own build; final grant report.

PROJECT TEAM

The project is led by Jim Rospopo, KE4CON, FieldComms architect and developer, supported by MCESV volunteer operators who conduct beta testing, net operations, and exercise deployments. MCESV operates under McHenry County Emergency Management Agency and runs weekly amateur and Starcom nets with an active membership roster. The v1.0 system, its documentation, and its GitHub repository demonstrate the team's capability to deliver.

BUDGET

All labor is volunteer — no personnel costs are requested. Equipment pricing was verified against vendor listings in June 2026; Raspberry Pi pricing reflects the current industry memory shortage.

ITEM	DETAIL	COST
FieldComms server	Raspberry Pi 5 16GB, Pironman MAX 5 enclosure, 2× 1TB NVMe (RAID 1), PSU, microSD	\$780
44Net gateway	Raspberry Pi 5 16GB, Argon NEO 5 case, 256GB M.2 SSD, PSU	\$400
Networking	3× ASUS RT-BE58 Go Wi-Fi 7 (primary + 2 AiMesh nodes), UniFi Switch Lite 16 PoE, CAT6 cabling	\$530
Cellular WAN	InstyConnect Drum omnidirectional antenna, Switchblade directional backup, spare PoE cable	\$820
Satellite WAN	Starlink kit, Ethernet adapter, USB-Ethernet adapter	\$545
Operator workstations	4× Raspberry Pi 500+ Desktop Kit (16 GB, 256 GB NVMe, mechanical keyboard, PSU, mouse, HDMI), 4× Raspberry Pi Monitor 15.6"	\$2,120
HF station	Icom IC-7300MK2 transceiver, 2× Astron RS-35M-AP 13.8V power supply (primary + spare), LDG IT-100 station autotuner	\$2,368
PACTOR HF modem	SCS PXdragon DR-9400 (current-generation PACTOR-4, replaces discontinued DR-7800) — PACTOR-1/-2/-3/-4, Packet Radio, Robust Packet — USB-C + Ethernet LAN standard — Winlink/Airmail/RMS Express — IC-7300MK2 interface cables	\$2,159
VHF/UHF station	Yaesu FTM-510DR, Comet CA-2X4SRNMO antenna, mag mount, Digirig Mobile + verified Yaesu FTM cable (\$20), Mobilinkd TNC4 + verified Yaesu FTM cable (\$25), USB-C data cable (\$10)	\$1,047
Field antenna system	Chameleon MPAS 2.0 modular portable antenna (160m-6m, deploys in under 2 minutes), CHA URT1 remote feedpoint tuner, 300 ft Chameleon antenna wire, 4 wire winders	\$1,350
Printing	Color laser MFP (Brother MFC-L3770CDW) for on-site IAP packages and ICS forms	\$400
Transport cases	6× Harbor Freight Apache 4800 XL weatherproof cases (IP65, interior 17-7/8×12-7/8×6-5/8", pick-and-pull foam) — \$59.99 list; coupons frequently available	\$360
Field power & accessories	UPS battery backup, GPS receiver, backup drives	\$530
Beta-year connectivity	InstyConnect data plan — standby 12 months + 3 exercise activations	\$400
Outreach	Hamfest table fees, printed replication guides and quick references, workshop materials	\$500
Contingency (~10%)	Shipping, taxes, cable ends, unforeseen substitutions	\$1,405
TOTAL REQUESTED (equipment + contingency)		\$15,439
Estimated tax + shipping (Woodstock IL 60098)	IL sales tax 8.25% + UPS/USPS ground estimates — see separate Tax & Shipping document	~\$1,376

ITEM	DETAIL	COST
TOTAL DELIVERED COST	Equipment + contingency + tax + shipping to 60098	~\$16,815

In-kind contribution: more than 800 volunteer development hours (complete working v1.0 software and 190+ pages of documentation), the developer-owned prototype used for continued development, and ongoing volunteer labor for testing, training, and outreach. ARDC is the sole funder being approached for this project.

REACH, IMPACT, AND MEASUREMENT

Direct: MCESV's active volunteer operators use the system at weekly amateur and Starcom nets (two nets per week, year-round) and at county activations and exercises, serving a county of roughly 310,000 residents.

Replication: the primary measure of success. Targets for the grant year: public v1.0 open-source release; two hamfest presentations and one ARES/RACES forum presentation reaching 200+ operators; one hands-on build workshop; at least one neighboring organization starting its own build; repository stars, forks, and clones tracked and reported. Every replicated system is also a new active 44Net gateway.

Model contribution: FieldComms fills a documented gap — no existing open-source project packages offline-first incident management, net logging with offline FCC data, ICS tooling, and 44Net integration into a club-buildable kit with professional documentation. AREDN provides the network layer; FieldComms provides the missing application layer, and the two are complementary.

SUSTAINABILITY

Hardware is owned by MCESV, a permanent 501(c)(3) organization, and has no recurring software costs — the platform is self-hosted with no licenses or subscriptions. The cellular plan idles at \$5/month (funded by the club) and activates only for exercises and incidents. The software is maintained in a public repository where the replication community contributes fixes and features. Weekly net use ensures the system and the operator skills stay current rather than sitting on a shelf.

COMMUNITY SUPPORT

Letters of support will accompany this application from McHenry County Emergency Management Agency and regional mutual-aid partners. The system was developed at the request of, and in continuous consultation with, MCESV net operators and county emergency management staff. [Attach letters before submission.]

OPEN ACCESS COMMITMENT

All software produced and released under this grant will carry the GNU Affero General Public License v3 (AGPLv3) — ARDC's preferred license — ensuring that all improvements, including hosted derivatives, remain open to the community; all documentation, guides, and video will carry CC-BY-SA 4.0. The repository at github.com/KE4CON/FieldComms-IMS will be public, and the replication guide will be written explicitly so that a club with no prior Linux experience can complete a build. No component of the system depends on proprietary services: FCC data is public domain, map tiles and radar are open government/academic sources, and all server software is open source.

Contact: Jim Rospopo, KE4CON · McHenry County Emergency Services Volunteers (K9ESV) · [email / phone / EIN to be added before submission] · Repository: github.com/KE4CON/FieldComms-IMS